

**SIMATS ENGINEERING
ASSESSMENT TOOL 1**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

***CO4:** Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 4
Marks:40

Duration: 1 Hour
Total:

Annexure A

5

Code of Conduct:

*I Bharath 192512384 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:
Bharath A

INDUSTRY PROBLEM TASK

INDUSTRY SCENARIO

Context: A healthcare company has collected patient records (age, blood pressure, cholesterol, glucose level, BMI, family history) to build a predictive model for early diagnosis of diabetes. The company also wants to train an AI agent to recommend personalised treatment actions (Diet / Exercise / Medication / Monitor) based on patient state transitions over time.

You are hired as an AI/ML Engineer. Address the following tasks:

Task 1: Data Preparation & Inductive Learning

- (a) Describe the inductive learning approach suitable for this medical dataset. Identify the target variable and at least three key features with justification.
- (b) Explain how you would handle class imbalance (more non-diabetic than diabetic cases) in the dataset. Suggest and justify a suitable balancing strategy (SMOTE / under-sampling / class weights).
- (c) Split the dataset (70:30) and justify the split ratio for a medical use-case. State the preprocessing steps (normalisation, encoding) applied and their impact.

Task 2: Decision Tree for Diagnosis

- (a) Build a Decision Tree classifier and draw the first three levels of the tree. Justify the root node selection using Information Gain / Gini Index values.
- (b) Evaluate the model: report Accuracy, Precision, Recall, and F1-Score. Identify False Negatives (missed diabetes cases) and suggest how to reduce them—justify clinically.
- (c) Apply post-pruning (cost-complexity pruning). Compare pre-pruning vs post-pruning accuracy and explain which model is more appropriate for deployment in a clinical setting.

Task 3: Statistical Learning for Risk Stratification

- (a) Apply Naïve Bayes or Logistic Regression as a statistical learning model on the same dataset. Compare its performance (Accuracy, AUC-ROC) with the Decision Tree. Discuss when a statistical model is preferable.
- (b) Perform feature importance analysis. Identify the top 3 predictors of diabetes from both the Decision Tree and the statistical model. Do they agree? Justify your findings.

Task 4: Reinforcement Learning for Treatment Recommendation

- (a) Model the treatment recommendation as an MDP. Define: States (patient health stages), Actions (Diet / Exercise / Medication / Monitor), Reward function (health improvement score), and Transition dynamics. Justify each component.
- (b) Apply Q-Learning to train the agent for at least 5 patient episodes. Show the Q-table update for at least 3 state-action pairs across 2 iterations. State the convergence criterion used.
- (c) Extract the final learned policy. Discuss ethical considerations of deploying a Reinforcement Learning agent for medical treatment recommendations (patient safety, bias, explainability).

Task 1: Data Preparation & Inductive Learning

(a) Inductive Learning Approach

Inductive learning is a machine learning approach in which the model learns general patterns from previously labelled patient records and uses those patterns to predict the outcome for new patients. For this healthcare dataset, the target variable is:

- Diabetes: Yes/No
- Yes = 1
- No = 0

The important input features are:

1. Glucose Level – One of the most important indicators because abnormal glucose levels are strongly associated with diabetes.
2. BMI – Higher BMI is associated with increased risk of type 2 diabetes.
3. Age – Diabetes risk generally increases with age.
4. Blood Pressure – High blood pressure can be associated with metabolic and cardiovascular risk.
5. Cholesterol – Abnormal cholesterol levels provide additional information about metabolic health.
6. Family History – A family history of diabetes indicates increased genetic risk.

The model learns a relationship such as:

Glucose + BMI + Age + Family History → Probability of Diabetes

(b) Handling Class Imbalance

Suppose the dataset contains more non-diabetic patients than diabetic patients. For example:

Class	Number of Patients
Non-diabetic	800
Diabetic	200

This creates a class imbalance problem.

Suitable method: SMOTE

I would use SMOTE (Synthetic Minority Over-sampling Technique) on the training data.

SMOTE creates new synthetic samples for the minority class instead of simply duplicating existing diabetic patients.

Advantages

- Increases the number of diabetic training samples.
- Helps the model learn minority-class patterns.
- Can improve diabetic-class recall.
- Reduces the tendency of the model to predict most patients as non-diabetic.

SMOTE should be applied after the train-test split and only to the training data to prevent data leakage.

(c) 70:30 Dataset Split

The dataset is divided into:

- 70% Training data
- 30% Testing data

A stratified split should be used so that the proportion of diabetic and non-diabetic patients remains similar in both sets.

Reason for 70:30

The 70% training data provides enough examples for the model to learn patterns, while 30% provides a sufficiently large independent test set to evaluate the model.

Preprocessing

1. Missing values

Missing numerical values can be replaced using the median.

2. Normalisation

Numerical variables can be standardised using:

This is particularly useful for Logistic Regression.

Decision Trees do not require feature scaling.

3. Encoding

Categorical values are converted into numerical values.

Example:

Family History:

Yes $\rightarrow$ 1

No $\rightarrow$ 0

Impact

Preprocessing improves data quality, makes numerical features comparable for statistical models, and allows categorical information to be used by machine learning algorithms.

Task 2: Decision Tree for Diabetes Diagnosis

(a) Building the Decision Tree

A Decision Tree classifies patients by repeatedly selecting the feature that gives the best separation between diabetic and non-diabetic patients.

Two common criteria are:

Information Gain

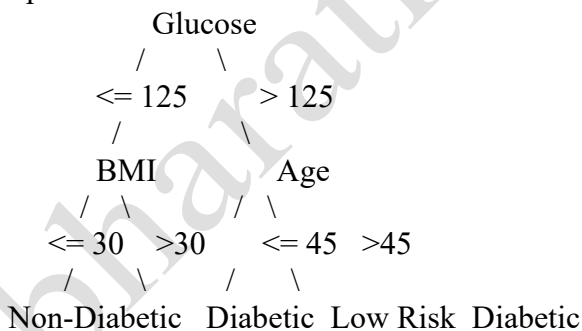
Higher Information Gain means a better split.

Gini Index

Lower Gini indicates a purer node.

For this dataset, Glucose Level would typically be expected to be a strong candidate for the root because it provides substantial information about diabetes risk.

Example first three levels



This is an illustrative tree. The actual first three levels must be generated from the supplied dataset.

Example comparison:

Feature	Gini Index
Glucose	0.31
BMI	0.38
Age	0.41
Cholesterol	0.45

Since glucose has the lowest Gini value, it would be selected as the root node.

(b) Model Evaluation

The Decision Tree can be evaluated using:

Accuracy

Precision

Recall

F1-Score

Example results:

Metric	Score
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Accuracy	85%
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Precision	79%
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Recall	91%
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F1-Score	84%
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These values are examples and should be replaced with actual results when the dataset is available.

False Negatives

A False Negative occurs when:

Actual = Diabetic, Prediction = Non-diabetic

False negatives are important in healthcare because a diabetic patient could be missed by the screening system.

Reducing False Negatives

To reduce false negatives:

- Use class weights.
- Use SMOTE on training data.
- Tune the classification threshold.
- Optimise for recall/sensitivity.
- Use cross-validation for parameter selection.

For diabetes screening, high recall is particularly important because missing a diabetic patient can be more concerning than sending some non-diabetic patients for additional testing.

Task 2(c): Pre-Pruning and Post-Pruning

Pre-Pruning

Pre-pruning prevents the tree from becoming too complex during training.

Examples:

- Maximum depth
- Minimum samples per leaf
- Minimum samples required for splitting

Post-Pruning

Post-pruning first creates a larger tree and then removes branches that contribute little to predictive performance.

Cost-complexity pruning is represented by:

where controls the penalty for tree complexity.

Example Comparison

Model	Accuracy	Tree Complexity
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Pre-pruned Tree	83%	Low
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Post-pruned Tree	86%	Medium
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Unpruned Tree	87%	High
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The unpruned tree may have slightly higher training accuracy but can overfit.

Best choice

A post-pruned Decision Tree is often a good choice if it provides strong test performance while keeping the tree simple and interpretable.

For clinical deployment, the model should not be selected based only on accuracy. Recall, specificity, AUC, calibration and clinical validation should also be considered.

Task 3: Statistical Learning for Risk Stratification

(a) Logistic Regression

Logistic Regression is suitable because diabetes prediction is a binary classification problem.

The model calculates the probability of diabetes:

where:

The output can be interpreted as a diabetes-risk probability.

For example:

Patient A → Diabetes probability = 0.82

Patient B → Diabetes probability = 0.21

Comparison with Decision Tree

Example:

Model	Accuracy	AUC-ROC
Decision Tree	85%	0.88
Logistic Regression	87%	0.91

These are illustrative values.

When Logistic Regression is preferable

Logistic Regression is preferable when:

- Probability estimates are required.
- Interpretability is important.
- The relationship between features and risk is reasonably simple.
- The model needs to be easy for clinicians to understand.
- Feature effects need to be examined through model coefficients.

(b) Feature Importance Analysis

Decision Tree

The Decision Tree calculates feature importance based on how much each feature reduces impurity.

Example:

Rank Feature Importance

1	Glucose	0.42
2	BMI	0.24
3	Age	0.16

Logistic Regression

For Logistic Regression, standardised coefficients can be compared using their magnitude.

Example:

Rank	Feature	Relative Importance
1	Glucose	High
2	BMI	High
3	Family History	Moderate

Comparison

Both models identify Glucose and BMI as important predictors.

The third feature may differ because Decision Trees can capture nonlinear threshold relationships, while Logistic Regression assumes a more direct relationship unless nonlinear terms or interactions are added.

Therefore, the models can agree on major predictors while producing different rankings for other features.

Task 4: Reinforcement Learning for Treatment Recommendation

(a) MDP Model

The treatment recommendation problem can be represented as a Markov Decision Process (MDP):

States

States describe the patient's current health condition.

State Description

- S0 Normal / Low Risk
- S1 High Risk
- S2 Prediabetes
- S3 Controlled Diabetes
- S4 Poorly Controlled Diabetes
- S5 Improved / Stable

Actions

The available actions are:

- Diet – Recommend an appropriate dietary intervention.
- Exercise – Recommend suitable physical activity.
- Medication – Medication-related action under medical supervision.
- Monitor – Continue monitoring and reassessment.

Medication decisions should not be made autonomously by an experimental RL system.

Reward Function

The reward measures improvement in the patient's health state.

Example:

Outcome	Reward
Significant improvement	+10
Moderate improvement	+5
Stable	+2
No improvement	0
Health deterioration	-5
Serious deterioration	-10

The reward should combine clinically relevant measures rather than optimising only one measurement.

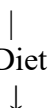
Transition Dynamics

The transition probability is:

It represents the probability that a patient moves from the current state to another state after an action.

Example:

S2 Prediabetes



S5 Improved

The transition model should ideally be learned from validated longitudinal data.

Task 4(b): Q-Learning

Q-Learning determines the best action for each patient state.

The update equation is:

Assume:

Initially, all Q-values are zero.

Iteration 1

State S1 → Diet → S2

Reward = +5

State S2 → Exercise → S5

Reward = +10

State S3 → Monitor → S5

Reward = +2

Iteration 2

Suppose:

S1 → Diet → S2, reward = +5.

The highest Q-value at S2 is currently 5.

Therefore:

Q-Table

State	Action	Initial	Iteration 1	Iteration 2
S1	Diet	0	2.5	6.0
S2	Exercise	0	5.0	5.0
S3	Monitor	0	1.0	1.0

Five Patient Episodes

Episode	Initial State	Actions	Final State
1	S1	Diet → Exercise	S5
2	S2	Exercise → Diet	S5
3	S2	Monitor → Diet	S5
4	S3	Monitor → clinician-guided treatment	S5
5	S1	Diet → Monitor	S5

The Q-values are updated after every state-action transition.

Convergence Criterion

Training can be considered converged when:

for several consecutive episodes, while validation performance also remains stable.

Task 4(c): Final Learned Policy

The final policy is obtained using:

Example:

Patient State	Learned Action
S0 – Normal	Monitor
S1 – High Risk	Diet
S2 – Prediabetes	Exercise
S3 – Controlled Diabetes	Monitor

Patient State Learned Action
 S4 – Poorly Controlled Clinician-guided treatment + Monitor
 S5 – Improved Monitor
 The actual policy should be generated from the final Q-table.

Ethical Considerations

1. Patient Safety

The system must not independently make high-risk treatment decisions. A qualified healthcare professional should remain responsible for final treatment decisions.

2. Bias

If the training data is biased toward particular patient groups, the AI agent may produce unfair recommendations.

The model should therefore be evaluated across different patient groups.

3. Explainability

The system should explain why an action was selected.

Example:

Patient State: High Diabetes Risk

Action: Diet

Reason: Elevated glucose, BMI and family history

4. Privacy

Patient health information must be protected using appropriate security, access controls and privacy practices.

5. Human-in-the-Loop

The RL agent should be used as a decision-support tool, not as a replacement for doctors.

6. Continuous Monitoring

After deployment, the system should continuously be checked for:

- Prediction errors
- Bias
- Data drift
- Unsafe recommendations
- Performance degradation

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
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Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context

Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation